

# Emergent Intelligence in Decentralized Swarms

*Capstone Research Draft (5–10 Agents)*

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## 1. Introduction

Emergent coordination in decentralized systems is a fundamental problem spanning multi-agent systems, robotics, and collective intelligence. Unlike centralized control, swarm systems rely on local interactions and shared incentives to produce globally coherent behavior.

This study investigates whether coordinated behavior can arise **without explicit communication or centralized control**, driven solely by environmental constraints and reward structures within a Multi-Agent Reinforcement Learning (MARL) framework.

This work positions reward structure as an implicit communication channel through which coordination emerges.

Emergence is defined as the statistically significant ( $p < 0.05$  across independent runs) improvement in global task performance accompanied by the formation of structured, coordinated behaviors not explicitly encoded in agent policies or communication mechanisms. Emergence is further validated through comparison with baseline heuristic and independent agent policies to ensure that observed coordination is not attributable to trivial strategies.

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## 2. Research Objective

The primary objective is to evaluate:

Under what conditions (reward structure and observability constraints) does decentralized coordination outperform independent agent policies?

We hypothesize that:

- Coordination efficiency increases with  $\alpha$ , potentially exhibiting non-linear or threshold behavior
- A critical threshold  $\alpha^*$  exists at which the system transitions from disordered to coordinated dynamics

- $\alpha^*$  is defined as the smallest value of  $\alpha$  for which coordination efficiency exceeds baseline performance with statistical significance ( $p < 0.05$ ).
  - Independent PPO exhibits reduced coordination stability under global reward structures due to non-stationarity introduced by independently updating policies.
  - Purely local reward signals lead to uncoordinated, competitive behavior
  - Performance is evaluated relative to independent agent baselines and heuristic policies to determine whether emergent coordination yields statistically significant improvements.
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## 3. System Architecture

### 3.1 Environment

- 2D spatial domain (grid-based or continuous)
- Partial observability via radius-limited perception
- Static or dynamically generated task layouts

To ensure analytical rigor, system behavior will be visualized using:

- Trajectory overlays
  - Spatial heatmaps of agent movement
  - Phase plots illustrating transition from disordered to structured motion
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### 3.2 Agents

- Population: 5–10 agents
  - Homogeneous policy (shared neural network)
  - No centralized controller or global state access
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### 3.3 State Representation

Each agent observes only local information:

- Positions and velocities of nearby agents
- Nearby environmental features (targets, obstacles)

To enforce symmetry:

- No agent identifiers are included in observations

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### 3.4 Action Space

- Discrete: {North, South, East, West}  
or
  - Continuous: velocity vector in 2D space
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## 4. Learning Framework

The system is implemented using:

- **PyTorch** for policy learning
  - **PettingZoo** for MARL environment design
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### 4.1 Algorithms

- Independent PPO (baseline)
  - Shared-policy PPO (primary approach)
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### 4.2 Reward Design

A multi-target foraging task is used to evaluate coordination.

#### Global Reward (Sparse)

- +1 if all targets are collected
- 0 otherwise

This configuration tests:

- Whether coordination can emerge **without shaping or guidance**
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#### Weakly Shaped Global Reward

- +1 per collected target (shared across agents)
- -0.01 per timestep

Provides a learning gradient while maintaining global alignment.

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### Local Reward (Self-Interested Agents)

- +1 when an individual agent collects a target

Expected outcome:

- This setting is expected to produce competitive, low-efficiency behavior
  - Reduced global efficiency
  - Potential clustering instability
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### Hybrid Reward

$$R = \alpha R_{\text{global}} + (1 - \alpha) R_{\text{local}}$$

- $\alpha \in [0,1]$  controls coordination pressure
  - Enables continuous analysis rather than discrete comparison
  - This formulation allows reward structure to function as an implicit coordination mechanism, enabling analysis of its role as a surrogate for communication.
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## 4.3 Key Design Constraint

No directional or distance-based shaping is used.

This ensures that:

- Coordination is not artificially induced
  - Observed behavior qualifies as **emergent**, not engineered
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## 5. Experimental Design

### 5.1 Independent Variables

- Reward structure (Global, Local, Hybrid  $\alpha$  sweep)
- Communication (None vs minimal signal)
- Learning algorithm (Independent vs Shared PPO)

- Episode ends when:
    - All targets collected OR
    - Max timestep reached
    - Episodes that reach the maximum timestep without task completion are considered failures.
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## 5.2 Controlled Variables

- Agent count (5–10)
  - Observation radius
  - Environment distribution
  - Training duration
  - $\geq 5$ –10 random seeds per condition
  - All experiments are conducted with fixed random seeds for reproducibility, in addition to randomized runs for statistical evaluation.
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## 5.3 Experimental Matrix

| ID | Reward Type | $\alpha$ | Communication | Algorithm       | Expected Behavior                               |
|----|-------------|----------|---------------|-----------------|-------------------------------------------------|
| E1 | Global      | —        | None          | Shared PPO      | Emergent coordination                           |
| E2 | Local       | —        | None          | Shared PPO      | uncoordinated,<br>competitive dynamics          |
| E3 | Hybrid      | 0.25     | None          | Shared PPO      | Weak coordination                               |
| E4 | Hybrid      | 0.5      | None          | Shared PPO      | Balanced behavior                               |
| E5 | Hybrid      | 0.75     | None          | Shared PPO      | Strong coordination                             |
| E6 | Global      | —        | None          | Independent PPO | Reduced stability and inconsistent coordination |
| E7 | Global      | —        | Minimal       | Shared PPO      | Faster convergence                              |

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## 5.4 Additional Conditions

- Observation noise (Gaussian perturbation)
  - Randomized spawn configurations
  - Removal of agent identity
  - Minimal communication (1 scalar signal)
  - All experiments are conducted under identical training budgets to ensure fair comparison across conditions.
  - Environmental configurations are sampled from a distribution to ensure generalization and prevent policy overfitting to fixed spatial patterns.
  - Each experiment is repeated across multiple random initializations to ensure robustness against stochastic training dynamics.
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## 6. Baseline Comparison

To validate emergent intelligence, learned policies are compared against:

- Greedy heuristic (nearest-target pursuit)
- Simple rule-based swarm behavior (flocking dynamics)
- These baselines establish whether learned coordination exceeds simple reactive or rule-based strategies.
- Baselines also serve to identify whether observed performance improvements arise from genuine coordination rather than unintended reward exploitation.

This ensures:

- Performance gains are not due to trivial strategies
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## 7. Evaluation Metrics

### Performance Metrics

- Task success rate
- Time to convergence
- Collision rate
- Coordination efficiency is defined as task success normalized by total agent movement cost.

- $CoordinationEfficiency = \frac{\text{success rate}}{\text{total distance traveled}}$
  - Total distance is computed as the sum of all agent movement over an episode.
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## Efficiency Metrics

- Distance traveled per completed task
  - Coordination efficiency
  - Statistical significance is evaluated using mean and standard deviation across runs, with appropriate hypothesis testing (e.g., t-tests) where applicable.
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## Behavioral Metrics

### Policy Entropy

- Measures exploration vs convergence

### Spatial Order Parameter

- Agent alignment and clustering
  - Variance in heading direction
  - Average inter-agent distance
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## Emergence Metrics

### Role Differentiation Score

- Tracks divergence in agent behavior
- Based on:
  - Movement patterns
  - Task contribution
  - Activity distribution

Clustering analysis (e.g., k-means) is used to detect:

- Spontaneous specialization
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## 8. Expected Results

- Coordination emerges under global and hybrid reward structures
- Local reward leads to inefficient, competitive dynamics
- A **phase transition** occurs as  $\alpha$  increases
- Variance across runs is expected to decrease as coordination stabilizes.

- Deviations from these expectations will be analyzed to identify failure modes and limitations of decentralized learning.
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## 9. Signature Result

The central result of this study is the **Emergence Curve**:

- X-axis: reward mixing parameter ( $\alpha$ )
- Y-axis: coordination efficiency

Expected outcome:

- Low  $\alpha \rightarrow$  disordered behavior
- High  $\alpha \rightarrow$  coordinated swarm behavior
- Critical transition region between the two
  - $\alpha^*$  (alpha-star)

Communication is expected to:

- Shift the transition threshold
- Improve convergence speed

The emergence curve characterizes a phase transition in system behavior, where increasing global reward influence induces a shift from disordered, independent actions to coordinated swarm dynamics. This transition is hypothesized to exhibit characteristics analogous to phase transitions in statistical systems, where small changes in reward structure induce large-scale behavioral reorganization. This result serves as the primary empirical validation of emergent coordination within the system.

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## 10. Robustness Analysis (Optional Extension)

Introduce agent failure during execution:

- Randomly disable agents mid-episode
  - % of agents removed (e.g., 10–30%)
  - Timing (early vs mid vs late episode)

Measure:

- Performance degradation
- Recovery time



This evaluates:

- Emergent resilience
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## **11. Deliverables**

- Trained multi-agent simulation
- Visualization suite (trajectories, heatmaps, phase plots)
- Comparative analysis across experimental conditions
- Emergence curve and supporting metrics